

SYSLOG

1. What is Syslog?

Syslog = System Logging Protocol

Syslog is an industry-standard protocol used for **message logging** on network devices such as routers, switches, firewalls, and servers.

It records events such as:

- Interface going UP/DOWN
- OSPF neighbor going UP/DOWN
- Configuration changes
- System restarts
- Errors and warnings
- Other important system events

Why is Syslog important?

Syslog is mainly used for:

- Monitoring
- Troubleshooting
- Finding the cause of incidents
- Keeping a record of important device events

Syslog messages can be:

1. Displayed in the CLI
2. Stored in the device's RAM
3. Sent to an external Syslog server

- Syslog is an industry standard protocol for message logging.
- On network devices, Syslog can be used to log events such as changes in interface status (up⇒ down), changes in OSPF neighbor status (up⇒ down), system restarts, etc.
- The messages can be displayed in the CLI, saved in the device's RAM, or sent to an external Syslog server.

```
R1(config)#int g0/0
R1(config-if)#no shutdown
R1(config-if)#
*Feb 11 03:02:55.304: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Feb 11 03:02:56.305: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

- Logs are essential when troubleshooting issues, examining the cause of incidents, etc.
- Syslog and SNMP are both used for monitoring and troubleshooting of devices. They are complementary, but their functionalities are different.



2. Why can I see Syslog messages even without configuring Syslog?

This was an important doubt.

Cisco IOS can display certain Syslog messages by default, especially on the **console**.

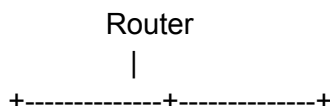
For example:

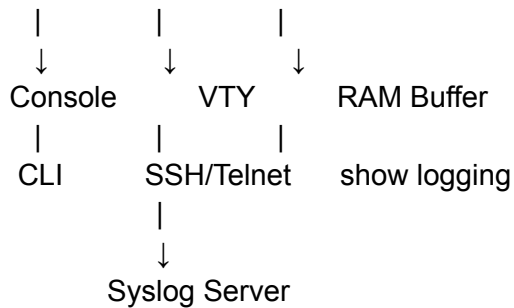
```
%LINK-3-UPDOWN: Interface GigabitEthernet0/0,
changed state to up
```

So:

Syslog is already functioning. The logging commands are used to CONTROL where the messages go and which severity levels are allowed.

For example:





Therefore, the commands are not necessarily "turning Syslog ON."

They are used to **control Syslog destinations, severity, storage, and formatting.**

3. Syslog Message Format

A Syslog message generally follows this format:

seq: timestamp: %facility-severity-MNEMONIC: description

Example:

```
000043: *Feb 11 05:06:43.331:
%SYS-5-CONFIG_I: Configured from console
```

Parts of the message

1. Sequence number

Indicates the order/sequence of messages.

Example:

```
000043
```

2. Timestamp

Shows when the event occurred.

Example:

Feb 11 05:06:43.331

3. Facility

Indicates which process/facility generated the message.

Example:

SYS
LINK
OSPF

4. Severity

Indicates how serious the event is.

Example:

5

5. Mnemonic

A short code describing the event.

Example:

CONFIG_I
UPDOWN
ADJCHG

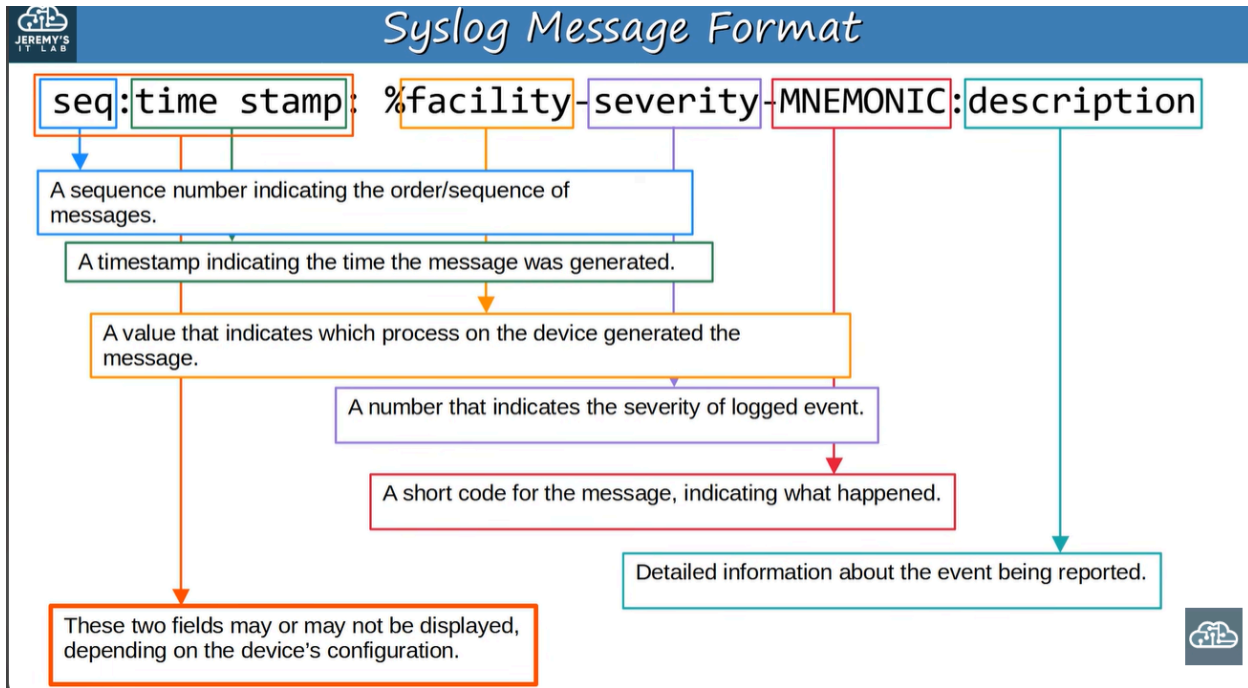
6. Description

Provides detailed information about what happened.

Example:

Configured from console by jeremy on console

Note: Sequence number and timestamp may or may not appear depending on configuration.



4. Syslog Severity Levels

Syslog has **8 severity levels: 0–7**.

Level	Keyword	Meaning
0	Emergency	System unusable
1	Alert	Immediate action required
2	Critical	Critical condition
3	Error	Error condition
4	Warning	Warning condition
5	Notice	Normal but significant condition
6	Informational	Informational message
7	Debugging	Debug-level message

Important rule

Lower number = higher severity

0 → Most serious

1

2

3

4

5

6

7 → Least serious



Syslog Severity Levels

Level	Keyword	Description
0	Emergency	System is unusable
1	Alert	Action must be taken immediately
2	Critical	Critical conditions
3	Error	Error conditions
4	Warning	Warning conditions
5	Notice	Normal but significant condition (Notification)
6	Informational	Informational messages
7	Debugging	Debug-level messages

Easy memory

0 Emergency

1 Alert

2 Critical

3 Error

4 Warning

5 Notice

6 Informational

7 Debugging

5. Understanding Severity Filtering

This is very important for the logging commands.

Suppose we configure:

logging console errors

errors = severity **3**.

It means:

Display severity **3 and MORE serious messages**.

Therefore:

0 Emergency	✓
1 Alert	✓
2 Critical	✓
3 Error	✓
4 Warning	✗
5 Notice	✗
6 Informational	✗
7 Debugging	✗

So **errors** does NOT mean only error messages.

It means **Error and everything more serious than Error**.

Similarly:

logging console warnings

means levels:

0, 1, 2, 3, 4

6. Syslog Message Examples

Example:

%LINK-3-UPDOWN:

Interface GigabitEthernet0/0, changed state to up

Breakdown:

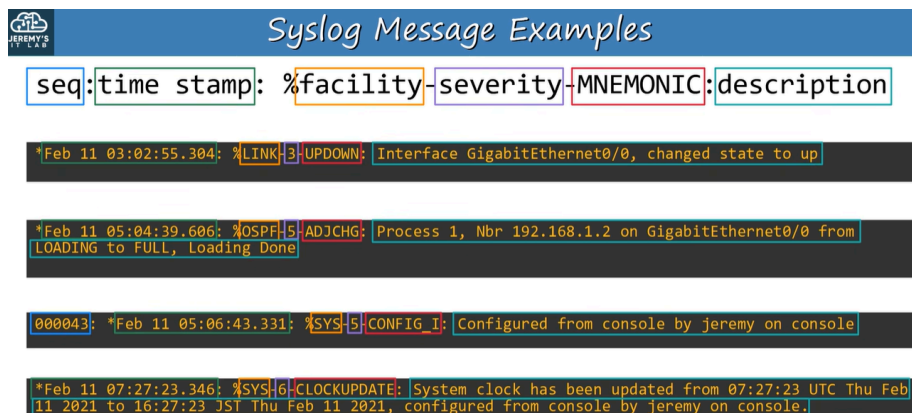
LINK → Facility
3 → Severity
UPDOWN → Mnemonic
Description → What happened

Another example:

%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.1.2 ... Loading Done

Here:

OSPF → Facility
5 → Severity
ADJCHG → Mnemonic



The diagram illustrates the structure of a Syslog message. It features a blue header bar with the Cisco logo and the text "Syslog Message Examples". Below the header, a general Syslog message format is shown: "seq:time stamp: %facility-severity-MNEMONIC:description". Each component is enclosed in a colored box: "seq:" (blue), "time stamp:" (green), "%facility" (orange), "-severity" (purple), "-MNEMONIC" (red), and ":description" (cyan). Below this, four example Syslog messages are displayed on a black background with colored text. Each example is broken down into its components with colored boxes: 1. "%Feb 11 03:02:55.304: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up" 2. "%Feb 11 05:04:39.606: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done" 3. "000043: *Feb 11 05:06:43.331: %SYS-5-CONFIG_I: Configured from console by jeremy on console" 4. "*Feb 11 07:27:23.346: %SYS-6-CLOCKUPDATE: System clock has been updated from 07:27:23 UTC Thu Feb 11 2021 to 16:27:23 JST Thu Feb 11 2021, configured from console by jeremy on console."

7. Syslog Logging Locations

Syslog messages can be sent to different locations.

1. Console

Messages are displayed when connected through the **console port**.

PC
|
| Console cable
↓
Router

Command:

logging console severity

2. VTY lines

Messages can be displayed when connected remotely using:

- SSH
- Telnet

PC
|
| SSH/Telnet
↓
Router

Command:

logging monitor severity

3. Buffer

Messages are stored in the router's **RAM**.

Command:

logging buffered size severity

View them using:

show logging

4. External Syslog Server

The router sends messages to a remote Syslog server.

Example:

Router —————→ Syslog Server
 192.168.1.100



Syslog Logging Locations

- **Console line:** Syslog messages will be displayed in the CLI when connected to the device via the console port. By default, all messages (level 0 – level 7) are displayed.
- **VTY lines:** Syslog messages will be displayed in the CLI when connected to the device via *Telnet/SSH* (coming in a later video). Disabled by default.
- **Buffer:** Syslog messages will be saved to RAM. By default, all messages (level 0 – level 7) are displayed.
→ You can view the messages with **show logging**.
- **External server:** You can configure the device to send Syslog messages to an external server.
*Syslog servers will listen for messages on **UDP port 514**. Remember that port number!

8. logging console severity

Command:

```
R1(config)# logging console severity
```

Example:

```
R1(config)# logging console 6
```

or:

R1(config)# logging console informational

This controls which Syslog messages are displayed on the **console line**.

If level 6 is configured:

0–6 → displayed

7 → not displayed

Important

The console may already display Syslog messages by default.

So this command is used to **control/filter the severity**, not simply to make Syslog exist.

Syslog Configuration

!configure logging to the console line
R1(config)#logging console 6
→ **logging console Level**
*you can use the level **number** (6) or **keyword** (informational)
*this will enable logging for the *informational* severity and higher

!configure logging to the vty lines
R1(config)#logging monitor informational
→ **logging monitor Level**
*same points as above about the level

!configure logging to the buffer
R1(config)#logging buffered 8192 6
→ **logging buffered [size] Level**
*same points as above about the level
*buffer size is in bytes

!configure logging to an external server
R1(config)#logging 192.168.1.100
R1(config)#logging host 192.168.1.100
→ **logging server-ip**
logging host server-ip
*these commands are the same!

R1(config)#logging trap debugging
→ **logging trap Level**
*same points as above about the level
*this sets the logging level for the external server

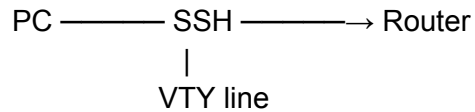
9. What is a VTY Line?

VTY = Virtual Teletype

A VTY line is a **virtual connection** used for **remote CLI access** to a Cisco device through:

- SSH
- Telnet

Example:



Console and VTY are different:

Console	VTY
Local access	Remote access
Console cable	SSH/Telnet
<code>line console 0</code>	<code>line vty 0 4</code>
Does not require network connectivity	Requires network connectivity

Example:

R1(config)# line vty 0 4

This enters the VTY line configuration.

10. logging monitor severity

This controls the severity of Syslog messages that can be displayed on **VTY sessions** such as SSH/Telnet.

Example:

R1(config)# logging monitor warnings

This allows severity:

0–4

to be displayed on VTY terminals.

Console vs VTY

logging console



Physical console connection

logging monitor



SSH/Telnet VTY connection

11. terminal monitor

Even if `logging monitor` is configured, Syslog messages are **not automatically displayed in an SSH/Telnet session**.

You must enable monitoring for your current terminal session:

R1# terminal monitor

Then Syslog messages can appear in your current SSH/Telnet window.

Important

logging monitor



Configures the VTY logging severity

terminal monitor



Enables log display in the current VTY session

`terminal monitor` is an **EXEC command**.



terminal monitor

- Even if **logging monitor** *Level* is enabled, by default Syslog messages will not be displayed when connected via Telnet or SSH.
 - For the messages to be displayed, you must use the following command:

```
R1#terminal monitor
```
 - This command must be used **every time you connect to the device via Telnet or SSH.**
-

12. logging buffered

Command:

logging buffered [size] severity

Example:

```
R1(config)# logging buffered 8192 6
```

This means:

8192 → buffer size in bytes

6 → severity level

The router stores the selected Syslog messages in its **RAM logging buffer**.

You can view them using:

```
R1# show logging
```

Example

```
logging buffered 4096 warnings
```

means:

4096 → buffer size

warnings → level 4

Therefore, levels:

0 Emergency	✓
1 Alert	✓
2 Critical	✓
3 Error	✓
4 Warning	✓
5 Notice	✗
6 Informational	✗
7 Debugging	✗

Important doubt clarification

If you configure:

logging buffered 4096 alerts

alerts = level 1.

Therefore only:

0 Emergency

1 Alert

are stored.

It does **not** include Warning.



logging synchronous

- By default, logging messages displayed in the CLI while you are in the middle of typing a command will result in something like this:

```
R1(config)#exit
R1#show ip in
*Feb 11 09:38:41.607: %SYS-5-CONFIG_I: Configured from console by jeremy on console
interface brief
```

- To prevent this, you should use the **logging synchronous** on the appropriate *line*. (I will talk more about 'line' configuration in the Telnet/SSH video!)

```
R1(config)#line console 0
R1(config-line)#logging synchronous
```

- This will cause a new line to be printed if your typing is interrupted by a message.

```
R1(config)#exit
R1#show ip int
*Feb 11 09:41:00.554: %SYS-5-CONFIG_I: Configured from console by jeremy on console
R1#show ip int
```

show ip int was reprinted on a new line. This makes it easier to continue typing the command.



13. logging server-ip vs logging host server-ip

These two forms can be seen in Cisco material:

logging 192.168.1.100

and:

logging host 192.168.1.100

They identify the remote Syslog server.

Example:

```
R1(config)# logging host 192.168.1.100
```

means:

Send Syslog messages to the Syslog server at **192.168.1.100**.

The word **host** here refers to the remote logging host/server.

Easy way to remember

logging host

↓

WHERE is the Syslog server?

192.168.1.100

The two forms shown in the training material are treated as equivalent ways of specifying the logging server.

[PASTE IMAGE 6 HERE – logging host/server command]

14. logging trap severity

Once we specify the Syslog server, we need to decide **which messages should be sent to it**.

Example:

```
R1(config)# logging host 192.168.1.100
```

```
R1(config)# logging trap warnings
```

logging host

Answers:

Where should I send the logs?

192.168.1.100

logging trap

Answers:

Which severity messages should I send?

For:

logging trap warnings

levels:

0–4

are sent to the external Syslog server.

Easy memory

logging host



WHERE?

logging trap



WHICH messages?

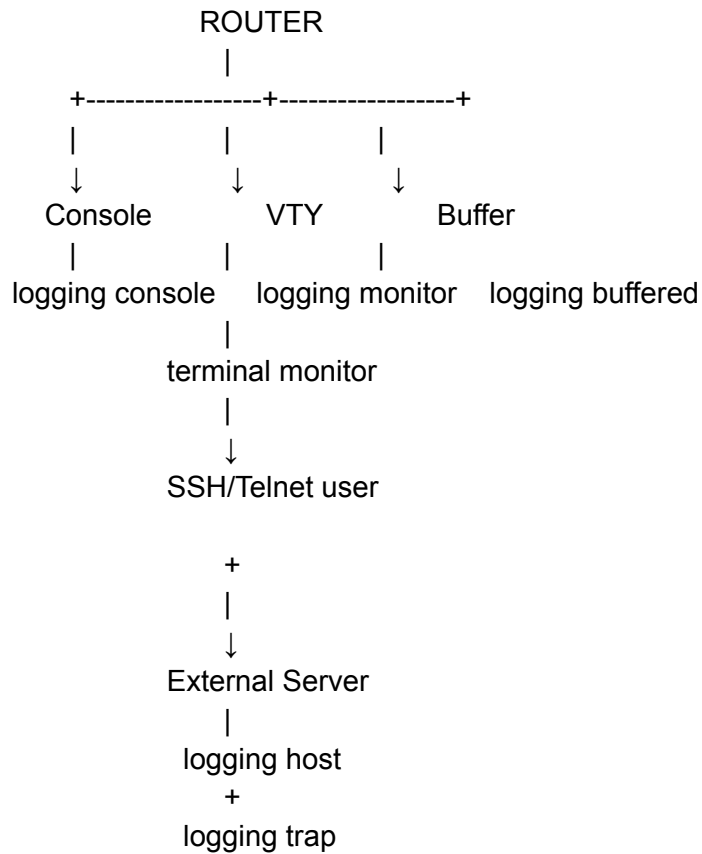
[PASTE IMAGE 6 HERE – logging trap command]

15. Complete Syslog Destination Comparison

Command	Destination	Purpose
<code>logging console severity</code>	Console	Controls console messages
<code>logging monitor severity</code>	VTY	Controls SSH/Telnet messages
<code>logging buffered size severity</code>	Router RAM	Stores logs in memory
<code>logging host server-ip</code>	External Syslog server	Specifies server

<code>logging trap severity</code>	External Syslog server	Controls severity sent to server
<code>terminal monitor</code>	Current VTY session	Displays logs in current SSH/Telnet session

Think of it like this:



16. logging synchronous



logging synchronous

- By default, logging messages displayed in the CLI while you are in the middle of typing a command will result in something like this:

```
R1(config)#exit
R1#show ip in
*Feb 11 09:38:41.607: %SYS-5-CONFIG_I: Configured from console by jeremy on console
interface brief
```

- To prevent this, you should use the **logging synchronous** on the appropriate *line*. (I will talk more about 'line' configuration in the Telnet/SSH video!)

```
R1(config)#line console 0
R1(config-line)#logging synchronous
```

- This will cause a new line to be printed if your typing is interrupted by a message.

```
R1(config)#exit
R1#show ip int
*Feb 11 09:41:00.554: %SYS-5-CONFIG_I: Configured from console by jeremy on console
R1#show ip int
```

show ip int was reprinted on a new line. This makes it easier to continue typing the command.



Problem:

Suppose you are typing:

R1# show ip interface brief

Suddenly a Syslog message appears:

%SYS-5-CONFIG_I: Configured from console...

It can interrupt your command and make the CLI difficult to read.

Solution

Configure:

```
R1(config)# line console 0
R1(config-line)# logging synchronous
```

Now when a Syslog message appears, the router moves the message to a separate line and **redispays your command prompt/command**, making it easier to continue typing.

Important

logging synchronous does **not stop** Syslog messages.

It only makes the CLI easier to use when messages appear.

[PASTE IMAGE 8 HERE – logging synchronous]

17. service timestamps log datetime

This adds the **date and time** to Syslog messages.

Command:

```
R1(config)# service timestamps log datetime
```

Example:

```
Feb 11 10:32:46:  
%SYS-5-CONFIG_I: Configured from console
```

This is useful because the administrator can know **when the event happened**.

18. service timestamps log uptime

Instead of the actual date/time, uptime shows how long the device has been running when the event occurred.

Command:

```
R1(config)# service timestamps log uptime
```

Conceptually:

```
00:15:32:  
%SYS-5-CONFIG_I: ...
```

Meaning the device had been running for approximately 15 minutes and 32 seconds when the event occurred.

Remember

datetime → actual date + time

uptime → how long device has been running



service timestamps / service sequence-numbers

```
R1(config)#service timestamps log ?
  datetime  Timestamp with date and time
  uptime    Timestamp with system uptime
  <cr>

R1(config)#service timestamps log datetime
R1(config)#
R1(config)#service sequence-numbers
R1(config)#exit
R1#
```

```
000039: *Feb 11 10:32:46: %SYS-5-CONFIG_I: Configured from console by
jeremy on console
```

datetime = timestamps will display the date/time when the event occurred.
uptime = timestamps will display how long the device had been running when the event occurred.

19. service sequence-numbers

This adds a sequence number to Syslog messages.

Command:

```
R1(config)# service sequence-numbers
```

Example:

```
000039: *Feb 11 10:32:46:
%SYS-5-CONFIG_I: Configured from console
```

Here:

000039

is the sequence number.

It helps identify the **order of messages/events**.

Example:

000039 → first message

000040 → next message

000041 → next message

Remember

Timestamp → WHEN did it happen?

Sequence number → IN WHAT ORDER did it happen?

[PASTE IMAGE 9 HERE]

20. Complete Syslog Configuration

Example:

```
R1(config)# logging console informational
```

```
R1(config)# logging monitor informational
```

```
R1(config)# logging buffered 8192 informational
```

```
R1(config)# logging host 192.168.1.100
```

```
R1(config)# logging trap debugging
```

Then, if using SSH:

```
R1# terminal monitor
```

For cleaner CLI output:

```
R1(config)# line console 0
```

```
R1(config-line)# logging synchronous
```

For timestamps:

```
R1(config)# service timestamps log datetime
```

For sequence numbers:

```
R1(config)# service sequence-numbers
```



Syslog Command Summary

```
R1(config)# logging console severity
R1(config)# logging monitor severity
R1(config)# logging buffered [size] severity
R1(config)# logging server-ip
R1(config)# logging host server-ip
R1(config)# logging trap severity
R1# terminal monitor
R1(config-line)# logging synchronous
R1(config)# service timestamps log [datetime | uptime]
R1(config)# service sequence-numbers
```



21. Why Do We Need All These Commands?

This was your main doubt.

You may already see:

```
%LINK-3-UPDOWN
```

without configuring anything.

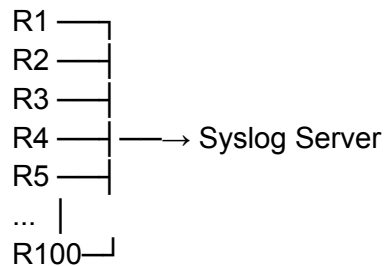
That's because Cisco can display Syslog messages by default.

But imagine a company with:

R1
R2
R3
R4
R5
...
R100

You cannot sit at every router's CLI and watch the messages.

So you configure a central Syslog server:



Now the administrator can collect logs centrally.

The commands allow us to control:

WHERE → Console / VTY / RAM / Server

WHAT → Severity level

WHEN → Timestamp

ORDER → Sequence number

HOW → Clean CLI display

22. Syslog Server and UDP Port

Syslog messages are commonly sent to a Syslog server using:

UDP port 514

Example:

Router
|
| Syslog
| UDP 514
↓
Syslog Server

[PASTE IMAGE 5 HERE – External Syslog Server and UDP 514]

23. Syslog vs SNMP

Both are used for network monitoring and troubleshooting, but their functions are different.

Syslog

Syslog is mainly used for **message/event logging**.

Examples:

- Interface went down
- OSPF neighbor changed
- Configuration changed
- Device restarted
- Error occurred

The device **sends the event message** to the Syslog server.

The server does not normally use Syslog to actively query the device for arbitrary information.

SNMP

SNMP is used to **retrieve and manage information from network devices**.

Examples:

- IP address
- Interface status
- CPU usage
- Temperature
- Other device statistics

SNMP can use:

GET → retrieve information

SET → modify certain variables

Easy difference

SYSLOG

Device → Server

"Something happened!"

SNMP

Server ↔ Device

"Tell me your status."

[PASTE IMAGE 11 HERE – Syslog vs SNMP]

24. Syslog vs SNMP – Exam Table

Feature	Syslog	SNMP
Main purpose	Event/message logging	Monitoring/management
Communication idea	Device sends logs	Manager queries device
Example	Interface went down	What is interface status?
GET	No	Yes
SET	No	Yes

Troubleshooting	Yes	Yes
Central server	Syslog server	SNMP manager
Common port	UDP 514	UDP 161

25. Important Commands – One-Line Revision

logging console severity

→ Controls Syslog messages displayed on the **console**.

logging monitor severity

→ Controls Syslog messages displayed on **VTY sessions**.

logging buffered size severity

→ Stores selected Syslog messages in the router's **RAM buffer**.

logging host server-ip

→ Specifies the **external Syslog server**.

logging trap severity

→ Controls which severity messages are sent to the **external Syslog server**.

terminal monitor

→ Displays Syslog messages in the **current SSH/Telnet VTY session**.

logging synchronous

→ Prevents Syslog messages from making CLI command entry difficult.

service timestamps log datetime

→ Adds the **date and time** to log messages.

service timestamps log uptime

→ Adds the **device uptime** to log messages.

service sequence-numbers

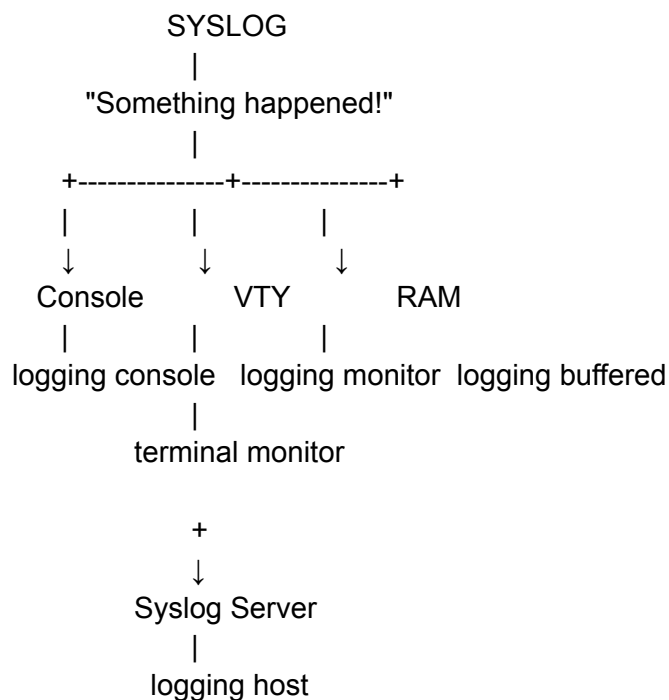
→ Adds **sequence numbers** to Syslog messages.

show logging

→ Displays the router's logging information/buffered logs.

26. Final Concept – Very Important

Remember Syslog as:



logging trap

And:

Severity:

0 = Most serious

7 = Least serious

Lower number = Higher severity

Most important distinction

logging console



WHAT appears on console?

logging monitor



WHAT appears on VTY?

logging buffered



WHAT is stored in RAM?

logging host



WHERE is the Syslog server?

logging trap



WHAT is sent to the Syslog server?

terminal monitor



SHOW logs in my current VTY session?

service timestamps



WHEN did it happen?

service sequence-numbers



IN WHAT ORDER did it happen?

logging synchronous



Keep my CLI readable while logs appear?

Quick Exam Definition

Syslog is an industry-standard message logging protocol used by network devices to record and transmit events such as interface changes, configuration changes, errors, and system events. Messages can be displayed on the CLI, stored in RAM, or sent to an external Syslog server for monitoring and troubleshooting.